

PIOLET

(Pioglitazone Tablets 15, 30 & 45mg)

Composition

PIOLET 15

Each tablet contains:

Pioglitazone Hydrochloride equivalent to Pioglitazone 15mg

PIOLET 30

Each tablet contains:

Pioglitazone Hydrochloride equivalent to Pioglitazone 30mg

PIOLET 45

Each tablet contains:

Pioglitazone Hydrochloride equivalent to Pioglitazone 45mg

Dosage form

Uncoated Tablet

Product Description

Piolet 15mg White to off white, round, biconvex, uncoated tablets debossed with 'P' on one side and '15' on other side.

Piolet 30mg White to off white, flat, round uncoated tablets with beveled edges debossed with 'PIO' on one side and '30' on the other side.

Piolet 45mg White to off white, flat, round uncoated tablets with beveled edges debossed with 'PIO' on one side and '45' on the other side.

Pharmacodynamics properties

Pharmacotherapeutic group: oral blood glucose lowering drugs;

Thiazolidinediones; ATC code: A10BG03.

Pioglitazone effects may be mediated by a reduction of insulin resistance. Pioglitazone appears to act via activation of specific nuclear receptors (peroxisome proliferator activated receptor gamma) leading to increased insulin sensitivity of liver, fat and skeletal muscle cells in animals. Treatment with pioglitazone has been shown to reduce hepatic glucose output and to increase peripheral glucose disposal in the case of insulin resistance. Fasting and postprandial glycaemic control is improved in patients with type 2 diabetes mellitus. The improved glycaemic control is associated with a reduction in both fasting and postprandial plasma insulin concentrations.

Pharmacokinetic properties

Absorption:

Following oral administration, pioglitazone is rapidly absorbed, and peak plasma concentrations of unchanged pioglitazone are usually achieved 2 hours after administration. Proportional increases of the plasma concentration were observed for doses from 2 – 60 mg. Steady state is achieved after 4–7 days of dosing. Repeated dosing does not result in accumulation of the compound or metabolites. Absorption is not influenced by food intake. Absolute bioavailability is greater than 80 %.

Distribution:

The estimated volume of distribution in humans is 0.25 l/kg. Pioglitazone and all active metabolites are extensively bound to plasma protein (> 99 %).

Metabolism:

Pioglitazone undergoes extensive hepatic metabolism by hydroxylation of aliphatic methylene groups. This is predominantly via cytochrome P450 2C8 although other isoforms may be involved to a lesser degree. Three of the six identified metabolites are active (M-II, M-III, and M-IV). When activity, concentrations and protein binding are taken into account, pioglitazone and metabolite M-III contribute equally to efficacy. On this basis M-IV contribution to efficacy is approximately three-fold that of pioglitazone, whilst the relative efficacy of M-II is minimal.

Elimination:

Following oral administration of radiolabelled pioglitazone to man, recovered label was mainly in faeces (55%) and a lesser amount in urine (45 %). In animals, only a small amount of unchanged pioglitazone can be detected in either urine or faeces. The mean plasma elimination half-life of unchanged pioglitazone in man is 5 to 6 hours and for its total active metabolites 16 to 23 hours.

Elderly:

Steady state pharmacokinetics are similar in patients age 65 and over and young subjects.

Patients with renal impairment:

In patients with renal impairment, plasma concentrations of pioglitazone and its metabolites are lower than those seen in subjects with normal renal function, but oral clearance of parent substance is similar. Thus free (unbound) pioglitazone concentration is unchanged.

Patients with hepatic impairment:

Total plasma concentration of pioglitazone is unchanged, but with an increased volume of distribution. Intrinsic clearance is therefore reduced, coupled with a higher unbound fraction of pioglitazone.

Indication/Usage

Pioglitazone is indicated in the treatment of type 2 diabetes mellitus:

as monotherapy

- in patients (particularly overweight patients) inadequately controlled by diet and exercise for whom metformin is inappropriate because of contraindications or intolerance

as dual oral therapy in combination with

- metformin, in patients (particularly overweight patients) with insufficient glycaemic control despite maximal tolerated dose of monotherapy with metformin

- a sulphonylurea, only in patients who show intolerance to metformin or for whom metformin is contraindicated, with insufficient glycaemic control despite maximal tolerated dose of monotherapy with a sulphonylurea.

Recommended Dose

Pioglitazone tablets are taken orally once daily with or without food. Pioglitazone may be initiated at 15mg or 30mg once daily. The dose may be increased in increments up to 45mg once daily. In combination with sulphonylurea, the current sulphonylurea dose can be continued upon initiation of pioglitazone therapy. If patients report hypoglycaemia, the dose of sulphonylurea should be decreased. In combination with metformin, the current metformin dose can be continued upon initiation of pioglitazone therapy.

Elderly:

No dosage adjustment is necessary for elderly patients (see section, Pharmacokinetic properties).

Patients with renal impairment:

No dosage adjustment is necessary in patients with impaired renal function (creatinine clearance > 4 ml/min) (see section, Pharmacokinetic properties). No information is available from dialysed patients therefore pioglitazone should not be used in such patients.

Patients with hepatic impairment:

Pioglitazone should not be used in patients with hepatic impairment (see section, Warnings & Precautions).

Children and adolescents:

There are no data available on the use of pioglitazone in patients under 18 years of age, and therefore its use is not recommended in this age group.

Route of Administration

Orally

Contraindication

Pioglitazone is contraindicated in patients with:

- hypersensitivity to the active substance or to any of the excipients
- cardiac failure or history of cardiac failure (NYHA stages I to IV)
- hepatic impairment

- diabetic ketoacidosis.
- Current bladder cancer or a history of bladder cancer

Warnings & Precautions

Fluid retention and cardiac failure:

Pioglitazone can cause fluid retention, which may exacerbate or precipitate heart failure. When treating patients who have at least one risk factor for development of congestive heart failure (e.g. prior myocardial infarction or symptomatic coronary artery disease), physicians should start with the lowest available dose and increase the dose gradually. Patients should be observed for signs and symptoms of heart failure, weight gain or oedema particularly those with reduced cardiac reserve. There have been cases of cardiac failure reported from the market when pioglitazone was used in combination with insulin or in patients with a history of cardiac failure; patients should be observed for signs and symptoms of heart failure, weight gain and oedema when pioglitazone is used in combination with insulin. Since insulin and pioglitazone are associated with fluid retention, concomitant administration may increase the risk of oedema. Pioglitazone should be discontinued if any deterioration in cardiac status occurs.

Bladder Cancer:

Cases of bladder cancer were reported more frequently in a meta-analysis of controlled clinical trials with pioglitazone (19 cases from 12506 patients, 0.15%) than in control groups (7 cases from 10212 patients, 0.07%) HR=2.64 (95% CI 1.11-6.31, P=0.029). After excluding patients in whom exposure to study drug was less than one year at the time of diagnosis of bladder cancer, there were 7 cases (0.06%) on pioglitazone and 2 cases (0.02%) in control groups. Available epidemiological data also suggest a small increased risk of bladder cancer in diabetic patients treated with pioglitazone in particular in patients treated for the longest durations and with the highest cumulative doses. A possible risk after short term treatment cannot be excluded.

Risk factors for bladder cancer should be assessed before initiating pioglitazone treatment (risks include age, smoking history, exposure to some occupational or chemotherapy agents e.g. cyclophosphamide or prior radiation treatment in the pelvic region). Any macroscopic haematuria should be investigated before starting pioglitazone therapy.

Patients should be advised to promptly seek the attention of their physician if macroscopic haematuria or other symptoms such as dysuria or urinary urgency develop during treatment.

Monitoring of liver function:

There have been rare reports of hepatocellular dysfunction during postmarketing experience (see section, Side effects). It is recommended, therefore, that patients treated with pioglitazone undergo periodic monitoring of liver enzymes. Liver enzymes should be checked prior to the initiation of therapy with pioglitazone in all patients. Therapy with pioglitazone should not be initiated in patients with increased baseline liver enzyme levels (ALT > 2.5 X upper limit of normal) or with any other evidence of liver disease.

Following initiation of therapy with pioglitazone, it is recommended that liver enzymes be monitored periodically based on clinical judgement. If ALT levels are increased to 3 X upper limit

of normal during pioglitazone therapy, liver enzyme levels should be reassessed as soon as possible. If ALT levels remain > 3 X the upper limit of normal, therapy should be discontinued. If any patient develops symptoms suggesting hepatic dysfunction, which may include unexplained nausea, vomiting, abdominal pain, fatigue, anorexia and/or dark urine, liver enzymes should be checked. The decision whether to continue the patient on therapy with pioglitazone should be guided by clinical judgement pending laboratory evaluations. If jaundice is observed, drug therapy should be discontinued.

Weight gain:

There was evidence of dose related weight gain, which may be due to fat accumulation and in some cases associated with fluid retention. In some cases weight increase may be a symptom of cardiac failure, therefore weight should be closely monitored. Part of the treatment of diabetes is dietary control. Patients should be advised to adhere strictly to a calorie-controlled diet.

Haematology:

There was a small reduction in mean haemoglobin and haematocrit. These changes primarily occurred within the first 4 to 12 weeks of therapy and remained relatively constant thereafter. These changes may be related to increased plasma volume and have rarely associated with any significant haematologic clinical effects.

Hypoglycaemia

As a consequence of increased insulin sensitivity, patients receiving pioglitazone in dual or triple oral therapy with a sulphonylurea or in dual therapy with insulin may be at risk for dose-related hypoglycaemia, and a reduction in the dose of the sulphonylurea or insulin may be necessary.

Eye disorders:

Post-marketing reports of new-onset or worsening diabetic macular oedema with decreased visual acuity have been reported with thiazolidinediones, including pioglitazone. Many of these patients reported concurrent peripheral oedema. It is unclear whether or not there is a direct association between pioglitazone and macular oedema but prescribers should be alert to the possibility of macular oedema if patients report disturbances in visual acuity; an appropriate ophthalmological referral should be considered.

Others:

The risk of fractures should be considered in the long term care of women treated with pioglitazone. As a consequence of enhancing insulin action, pioglitazone treatment in patients with polycystic ovarian syndrome may result in resumption of ovulation. These patients may be at risk of pregnancy. Patients should be aware of the risk of pregnancy and if a patient wishes to become pregnant or if pregnancy occurs, the treatment should be discontinued (see section, Pregnancy and Lactation).

Pioglitazone should be used with caution during concomitant administration of cytochrome P450 2C8 inhibitors (e.g. gemfibrozil) or inducers (e.g. rifampicin). Glycaemic control should be monitored closely. Pioglitazone dose adjustment within the recommended posology or changes in diabetic treatment should be considered (see section, Interaction with other medicaments).

Pioglitazone tablets contain lactose monohydrate and therefore should not be administered to patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption.

Interaction with other medicaments

Interaction studies have shown that pioglitazone has no relevant effect on either the pharmacokinetics or pharmacodynamics of digoxin, warfarin, phenprocoumon and metformin. Co-administration of pioglitazone with sulphonylureas does not appear to affect the pharmacokinetics of the sulphonylurea.

Pregnancy and Lactation

Use in pregnancy:

There are no adequate human data to determine the safety of pioglitazone during pregnancy. Foetal growth restriction was apparent in animal studies with pioglitazone. This was attributable to the action of pioglitazone in diminishing the maternal hyperinsulinaemia and increased insulin resistance that occurs during pregnancy thereby reducing the availability of metabolic substrates for foetal growth. The relevance of such a mechanism in humans is unclear and pioglitazone should not be used in pregnancy.

Use in breast-feeding:

Pioglitazone has been shown to be present in the milk of lactating rats. It is not known whether pioglitazone is secreted in human milk. Therefore, pioglitazone should not be administered to breast feeding women.

Side effects/Adverse Reactions

Adverse reactions reported in excess > 0.5 %) of placebo and as more than an isolated case in patients receiving pioglitazone in double-blind studies are listed below as MedDRA preferred term by system organ class and absolute frequency. Frequencies are defined as: very common > 1/10; common > 1/100, < 1/10; uncommon > 1/1000, < 1/100; rare > 1/10000, < 1/1000; very rare < 1/10000; not known (cannot be estimated from the available data). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

PIOGLITAZONE MONOTHERAPY

Eye disorders

Common: visual disturbance

Infection and infestations

Common: upper respiratory tract infection

Uncommon: sinusitis

Neoplasms benign, malignant and unspecified (including cysts and polyps)

Uncommon: Bladder Cancer

Investigations

Common: weight increased

Nervous system disorders

Common: hypoaesthesia

Uncommon: insomnia

PIOGLITAZONE IN COMBINATION THERAPY WITH METFORMIN

Blood and lymphatic system disorders

Common: anaemia

Eye disorders

Common: visual disturbance

Neoplasms benign, malignant and unspecified (including cysts and polyps)

Uncommon: Bladder cancer

Gastrointestinal disorders

Uncommon: flatulence

Investigations

Common: weight increased

Musculoskeletal system and connective tissue disorders

Common: arthralgia

Nervous system disorders

Common: headache

Renal and urinary disorders

Common: haematuria

Reproductive system and breast disorders

Common: erectile dysfunction

PIOGLITAZONE IN COMBINATION THERAPY WITH SULPHONYLUREA

Ear and labyrinth disorders

Uncommon: vertigo

Eye disorders

Uncommon: visual disturbance

Neoplasms benign, malignant and unspecified (including cysts and polyps)

Uncommon: Bladder cancer

Gastrointestinal disorders

Common: flatulence

General disorders and administration site conditions

Uncommon: fatigue

Investigations

Common: weight increased

Uncommon: increased lactic dehydrogenase

Metabolism and nutritional disorders

Uncommon: appetite increased, hypoglycaemia

Nervous system disorders

Common: dizziness

Uncommon: headache

Renal and urinary disorders

Uncommon: glycosuria, proteinuria

Skin and subcutaneous tissue disorders

Uncommon: sweating

Signs & Symptoms of overdose and Treatment

Patients have taken pioglitazone at higher than the recommended highest dose of 45 mg daily. The maximum reported dose of 120 mg/day for four days, then 180 mg/day for seven days was not associated with any symptoms.

Hypoglycaemia may occur in combination with sulphonylureas or insulin. Symptomatic and general supportive measures should be taken in case of overdose.

Storage

Do not store above 30°C.

Package

Pioglitazone Hydrochloride Tablets are available in pack sizes of 14 tablets (2 x 14T).

Shelf life

36 months

Piolet 15mg & 30mg:**Manufactured by:**

INTAS PHARMACEUTICALS LTD.

Camp road, Selaqui-248197,
Dehradun, Uttarakhand (India).

Piolet 45mg:

Manufactured by:

Intas Pharmaceuticals Ltd.

Matoda 382 210, Dist. Ahmedabad, India.

Product Registration Holder

JETPHARMA SDN BHD

No.13 Jalan Rajawali 2, Bandar
Puchong Jaya 47100 Puchong,
Selangor, Malaysia

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